

PAPER CODE: BCE-C611
SEMESTER EXAMINATION (MAY-2025)
CLASS: B.TECH
SEMESTER: VI
SUBJECT: CSE
PAPER TITLE: DISTRIBUTED SYSTEMS

2263 01202

Time: 3 Hours

Max. Marks: 70

Note: Question Paper is divided into two sections: **A and B**. Attempt both the sections as per given instructions.

SECTION-A (SHORT ANSWER TYPE QUESTIONS)

CO **BL**

Instructions: Answer any *five* questions in about 150 words each. Each question carries six marks.
(5 x 6 = 30 Marks)

Q-1	Explain about Distributed Systems? Explain their key characteristics with examples.	CO1	BL1
Q-2	Define Events and Notifications in a distributed system? Explain with examples.	CO2	BL2
Q-3	Explain the concept of Distributed Shared Memory (DSM) in distributed systems. What are the key design and implementation challenges associated with DSM?	CO1	BL1
Q-4	Explain the terms: 1) Replication 2) Consistency 3) Concurrency.	CO3	BL3
Q-5	Define Distributed Objects and Remote Invocation? Explain their importance in distributed systems.	CO2	BL1
Q-6	Explain the role of the Operating System in Distributed Computing?	CO2	BL2
Q-7	Explain about: 1) Race Condition 2) Context Switching 3) SOA.	CO4	BL2
Q-8	Explain the significance of global states in distributed systems and their role in debugging, deadlock detection, and fault tolerance.	CO1	BL2
Q-9	Provide a case study on Interprocess Communication (IPC) in UNIX, comparing pipes, message queues, shared memory, and sockets, including security and efficiency trade-offs.	CO4	BL4
Q-10	Discuss the Common Object Request Broker Architecture (CORBA) and its role in distributed computing.	CO3	BL3

SECTION-B (LONG ANSWER TYPE QUESTIONS)

CO **BL**

Instructions: Answer any *four* questions in detail. Each question carries 10 marks.

(4 x 10 = Marks)

Q-1	Discuss the concept of resource sharing in distributed systems, focusing on the Web. How do middleware, caching, and load balancing contribute to efficient resource management?	CO3	BL4
Q-2	Explain the system models used in distributed systems. Discuss the differences between three tier models and peer-to-peer models with examples.	CO2	BL2

Q-3	Discuss algorithms for distributed mutual exclusion and leader election, comparing Ricart-Agrawala, Maekawa's algorithms with their advantages and drawbacks.	CO3	BL2
✓ Q-4	Explain the role of events, and process states in distributed systems and their impact on consistency and coordination through time synchronization.	CO2	BL3
✓ Q-5	Explain the role of Name Services in distributed computing and how DNS ensures fault tolerance and load balancing in large-scale networks	CO1	BL1
Q-6	Discuss File Service Architecture in distributed systems, comparing stateful and stateless file services in terms of performance, reliability, and scalability.	CO4	BL2
✓ Q-7	Explain the concept of marshalling in distributed systems. How does it facilitate communication between heterogeneous systems, and what are the challenges associated with marshalling and unmarshalling data?	CO1	BL1
✓ Q-8	Discuss the importance of physical clock synchronization in distributed systems and compare Cristian's algorithm, Berkeley's algorithm, and NTP.	CO2	BL3
